# Supplementary Material

Experimental and DFT-Assisted Raman Characterisation of Isolated N-Acetylmuramic Acid

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## S1. Sample

N-acetylmuramic acid, Sigma-Aldrich A3007, CAS 10597-89-4, synthetic, ≥98% (TLC), white powder, MW 293.27 (anhydrous $C_{11}H_{19}NO_8$), stated residual methanol ≤1 mol/mol, stored 2–8 °C. The anomeric configuration is not specified by the supplier; the material is an anomeric mixture.

## S2. Instrument

B&W Tek i-Raman Plus, model BWS465-785H, spectrometer BTC665N-785H-SYS, with a BAC102-785E Raman microscope and 20× Plan objective. Excitation 785 nm (784.92 nm reported), nominal maximum output 495 mW, Class 3B. Detector 2048 pixels, range −46 to 2842 cm$^{-1}$, sampling 1.76 cm$^{-1}$/pixel at 400 cm$^{-1}$ and 1.43 cm$^{-1}$/pixel at 1800 cm$^{-1}$.

## S3. Measurement inventory

| Group | Spectra | Purpose |
|---|---|---|
| Acquisition-parameter survey | 98 | 5 integration times × 16 power settings |
| Replicate sets (9) | 104 | fixed conditions, fresh spot each |
| Photostability series | 10 | 90%, 30 s, same spot |
| Laser-damage series | 10 | 90%, 60 s, excluded from analysis |
| Substrate blanks | 5 | empty glass slide |
| Total | 227 | |

The reference spectrum uses 22 spectra from two replicate sets (70% and 90% power, 10 s, 5 accumulations). Weak bands were verified against 11 spectra at 30 and 60 s.

## S4. Computational details

Gaussian 16, route section:

```
# opt=(calcfc,tight) freq=raman b3lyp/6-311++g(d,p)
empiricaldispersion=gd3bj geom=connectivity int=ultrafine scf=(tight,xqc)
```

Starting geometry: $\beta$ anomer, PubChem CID 5462244. Normal termination, stationary point found, 111 normal modes, none imaginary. Scaling 0.980 below 1800 cm$^{-1}$, 0.967 above.

## S5. Optimised Cartesian coordinates

## S6. All calculated normal modes

Table S1: Optimised Cartesian coordinates (Å).

| # | El. | $x$ | $y$ | $z$ | # | El. | $x$ | $y$ | $z$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | O | -2.680794 | 0.670787 | -0.094904 | 21 | H | 0.115079 | -0.742934 | -0.965928 |
| 2 | O | 1.069463 | -0.759373 | 0.872185 | 22 | H | -0.461724 | 1.360374 | 1.192032 |
| 3 | O | -1.067922 | -2.615191 | 0.317311 | 23 | H | -1.448390 | -1.028364 | 1.618181 |
| 4 | O | -1.840721 | 2.758593 | -0.464037 | 24 | H | -2.405640 | -1.001894 | -1.294143 |
| 5 | O | -4.955878 | -0.860424 | -0.465147 | 25 | H | -1.460230 | 1.142196 | -1.687055 |
| 6 | O | 0.942881 | 3.225032 | 1.408684 | 26 | H | -3.949603 | -1.118455 | 1.337915 |
| 7 | O | 3.357269 | -2.528753 | -1.283455 | 27 | H | -3.830150 | -2.425509 | 0.133294 |
| 8 | O | 2.634316 | -0.412110 | -1.416760 | 28 | H | 1.474733 | 1.235877 | -0.949221 |
| 9 | N | 0.859442 | 1.760669 | -0.340525 | 29 | H | 1.285482 | -2.711805 | 0.221839 |
| 10 | C | -0.072282 | -0.460237 | 0.078650 | 30 | H | -1.622947 | -3.135841 | 0.904684 |
| 11 | C | -0.301541 | 1.055014 | 0.154576 | 31 | H | -2.582451 | 2.996868 | -1.030119 |
| 12 | C | -1.307206 | -1.222243 | 0.548677 | 32 | H | 3.458064 | -1.245169 | 1.811664 |
| 13 | C | -2.537918 | -0.741338 | -0.233338 | 33 | H | 3.527990 | -2.951307 | 1.307290 |
| 14 | C | -1.575401 | 1.409778 | -0.622512 | 34 | H | 2.293590 | -2.407382 | 2.465753 |
| 15 | C | -3.843175 | -1.342297 | 0.266764 | 35 | H | -4.912686 | 0.102812 | -0.432622 |
| 16 | C | 1.889137 | -1.825745 | 0.442405 | 36 | H | 3.455207 | 3.418734 | 0.666429 |
| 17 | C | 1.444020 | 2.757941 | 0.401236 | 37 | H | 3.199299 | 2.597780 | -0.898148 |
| 18 | C | 2.863583 | -2.130825 | 1.578298 | 38 | H | 2.581649 | 4.240235 | -0.627046 |
| 19 | C | 2.646027 | -1.473067 | -0.837832 | 39 | H | 3.815808 | -2.253317 | -2.092008 |
| 20 | C | 2.758411 | 3.268675 | -0.158613 | | | | | |

Table S2: All 111 calculated modes. $\nu$: unscaled; $\nu_s$: scaled; $S$: Raman activity (Å$^4$/amu); $A$: IR intensity (km/mol).

| # | $\nu$ | $\nu_s$ | $S$ | $A$ | # | $\nu$ | $\nu_s$ | $S$ | $A$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 34.82 | 34.13 | 0.878 | 0.24 | 57 | 1113.10 | 1090.84 | 4.810 | 26.71 |
| 2 | 40.87 | 40.05 | 1.180 | 1.29 | 58 | 1130.03 | 1107.43 | 6.554 | 236.52 |
| 3 | 42.12 | 41.27 | 0.253 | 0.85 | 59 | 1139.36 | 1116.58 | 3.055 | 276.09 |
| 4 | 57.31 | 56.16 | 0.221 | 0.30 | 60 | 1147.58 | 1124.63 | 2.746 | 103.06 |
| 5 | 60.11 | 58.90 | 0.466 | 1.02 | 61 | 1158.55 | 1135.38 | 1.888 | 35.56 |
| 6 | 71.46 | 70.03 | 0.400 | 2.22 | 62 | 1179.81 | 1156.21 | 3.271 | 19.03 |
| 7 | 84.59 | 82.90 | 1.031 | 8.94 | 63 | 1196.36 | 1172.43 | 4.411 | 9.66 |
| 8 | 90.70 | 88.88 | 0.103 | 2.52 | 64 | 1216.46 | 1192.13 | 2.990 | 0.21 |
| 9 | 107.93 | 105.77 | 0.123 | 3.53 | 65 | 1232.88 | 1208.22 | 2.125 | 38.71 |
| 10 | 120.61 | 118.20 | 0.119 | 4.76 | 66 | 1254.46 | 1229.37 | 0.970 | 5.11 |
| 11 | 148.73 | 145.75 | 0.087 | 4.91 | 67 | 1267.63 | 1242.27 | 3.847 | 11.20 |
| 12 | 175.55 | 172.04 | 0.847 | 3.11 | 68 | 1282.76 | 1257.10 | 6.022 | 125.07 |
| 13 | 182.62 | 178.97 | 0.081 | 3.27 | 69 | 1310.62 | 1284.41 | 1.484 | 150.91 |
| 14 | 213.68 | 209.41 | 0.203 | 0.24 | 70 | 1313.88 | 1287.60 | 4.543 | 16.90 |
| 15 | 230.14 | 225.54 | 0.284 | 56.07 | 71 | 1332.86 | 1306.21 | 1.307 | 4.22 |
| 16 | 241.59 | 236.76 | 1.744 | 50.31 | 72 | 1340.36 | 1313.55 | 2.719 | 12.40 |
| 17 | 244.28 | 239.40 | 1.135 | 0.86 | 73 | 1351.86 | 1324.82 | 0.715 | 4.98 |
| 18 | 264.14 | 258.86 | 1.077 | 13.63 | 74 | 1361.58 | 1334.35 | 1.938 | 2.20 |
| 19 | 285.05 | 279.35 | 0.571 | 23.88 | 75 | 1373.15 | 1345.69 | 5.360 | 4.35 |
| 20 | 294.95 | 289.05 | 3.245 | 1.37 | 76 | 1384.29 | 1356.60 | 2.107 | 5.58 |
| 21 | 301.70 | 295.66 | 2.174 | 73.20 | 77 | 1387.00 | 1359.26 | 10.986 | 10.60 |
| 22 | 314.12 | 307.84 | 1.561 | 12.38 | 78 | 1396.66 | 1368.73 | 2.703 | 37.89 |
| 23 | 340.59 | 333.78 | 0.949 | 3.77 | 79 | 1400.65 | 1372.63 | 2.211 | 23.11 |
| 24 | 345.33 | 338.42 | 1.337 | 21.34 | 80 | 1407.31 | 1379.17 | 3.795 | 4.71 |
| 25 | 385.45 | 377.74 | 1.774 | 8.89 | 81 | 1415.92 | 1387.61 | 7.768 | 8.86 |

| # | $\nu$ | $\nu_s$ | $S$ | $A$ | # | $\nu$ | $\nu_s$ | $S$ | $A$ |
|---|---|---|---|---|---|---|---|---|---|
| 26 | 411.66 | 403.43 | 3.917 | 25.26 | 82 | 1436.72 | 1407.98 | 5.368 | 48.93 |
| 27 | 423.89 | 415.42 | 1.354 | 38.83 | 83 | 1438.71 | 1409.93 | 3.856 | 9.26 |
| 28 | 449.53 | 440.54 | 0.633 | 81.10 | 84 | 1462.43 | 1433.18 | 2.650 | 5.42 |
| 29 | 469.85 | 460.45 | 1.027 | 32.68 | 85 | 1473.40 | 1443.94 | 5.230 | 11.66 |
| 30 | 504.26 | 494.17 | 2.844 | 19.54 | 86 | 1489.54 | 1459.75 | 7.552 | 15.05 |
| 31 | 523.85 | 513.37 | 2.287 | 10.96 | 87 | 1489.73 | 1459.94 | 5.450 | 3.78 |
| 32 | 566.96 | 555.62 | 3.176 | 3.86 | 88 | 1500.82 | 1470.80 | 6.370 | 1.81 |
| 33 | 581.59 | 569.96 | 2.284 | 9.63 | 89 | 1501.81 | 1471.77 | 4.275 | 7.42 |
| 34 | 584.97 | 573.27 | 1.838 | 93.18 | 90 | 1566.57 | 1535.24 | 2.154 | 257.10 |
| 35 | 594.02 | 582.14 | 0.910 | 55.28 | 91 | 1757.98 | 1722.82 | 12.805 | 308.61 |
| 36 | 620.81 | 608.39 | 0.625 | 18.43 | 92 | 1795.14 | 1759.24 | 10.169 | 328.18 |
| 37 | 632.03 | 619.39 | 0.459 | 11.06 | 93 | 2934.52 | 2837.68 | 48.614 | 47.10 |
| 38 | 639.48 | 626.69 | 0.830 | 24.68 | 94 | 2977.52 | 2879.27 | 139.250 | 7.80 |
| 39 | 653.21 | 640.15 | 1.868 | 57.36 | 95 | 2990.04 | 2891.37 | 11.131 | 51.84 |
| 40 | 706.31 | 692.19 | 1.470 | 36.92 | 96 | 3001.66 | 2902.61 | 83.689 | 16.66 |
| 41 | 746.20 | 731.27 | 5.290 | 26.55 | 97 | 3033.60 | 2933.49 | 24.378 | 35.48 |
| 42 | 809.38 | 793.19 | 7.406 | 9.66 | 98 | 3042.37 | 2941.98 | 181.917 | 7.57 |
| 43 | 890.78 | 872.97 | 3.213 | 22.88 | 99 | 3047.82 | 2947.24 | 190.231 | 15.33 |
| 44 | 923.28 | 904.81 | 2.333 | 3.01 | 100 | 3052.20 | 2951.48 | 22.588 | 11.52 |
| 45 | 945.58 | 926.67 | 3.568 | 6.15 | 101 | 3076.75 | 2975.22 | 31.609 | 12.31 |
| 46 | 999.46 | 979.47 | 2.644 | 8.69 | 102 | 3091.05 | 2989.04 | 72.808 | 23.83 |
| 47 | 1020.26 | 999.85 | 2.380 | 10.16 | 103 | 3108.35 | 3005.77 | 59.045 | 11.25 |
| 48 | 1026.75 | 1006.22 | 1.967 | 92.29 | 104 | 3122.14 | 3019.11 | 49.657 | 12.08 |
| 49 | 1042.37 | 1021.52 | 2.308 | 10.64 | 105 | 3129.29 | 3026.02 | 58.738 | 10.63 |
| 50 | 1049.19 | 1028.21 | 0.890 | 118.98 | 106 | 3135.97 | 3032.48 | 65.276 | 10.69 |
| 51 | 1056.02 | 1034.90 | 0.202 | 23.67 | 107 | 3565.54 | 3447.87 | 84.951 | 176.05 |
| 52 | 1063.47 | 1042.20 | 7.653 | 72.50 | 108 | 3752.61 | 3628.77 | 147.673 | 80.33 |
| 53 | 1076.34 | 1054.82 | 1.783 | 39.46 | 109 | 3804.46 | 3678.91 | 46.110 | 39.45 |
| 54 | 1089.60 | 1067.80 | 0.588 | 205.57 | 110 | 3830.63 | 3704.22 | 146.915 | 47.43 |
| 55 | 1094.55 | 1072.66 | 12.446 | 5.23 | 111 | 3853.40 | 3726.23 | 113.263 | 45.54 |
| 56 | 1101.87 | 1079.83 | 0.618 | 32.85 | | | | | |

## S7. Data files

- `Experimental_reference_spectrum.csv` — mean and standard deviation
- `Experimental_peak_list.csv` — 41 bands with confidence metrics
- `Experimental_vs_DFT_comparison.csv` — Table 1 as data
- `NAM_calculated_modes_all_111.csv` — full mode list
- `DFT_simulated_spectrum.csv` — broadened simulated spectrum